

Resistance of a Wire:  $R[\text{Ohms}] = \frac{\rho[\text{ohm-meters}]L[\text{meters}]}{A[\text{meters}^2]}$

Where  $A = \pi \left(\frac{d}{2}\right)^2$  if d in meters, or  $A_{CM} = (d_{mils})^2$

Resistivity of Copper Wire at 20C :  $1.68 * 10^{-8} \Omega \text{m}$

Resistance of a Conductor at Varying Temperature:

$R = R_{\text{ref}}[1 + \alpha(T - T_{\text{ref}})]$  where  $\alpha$  is located in T3.6

Mesh Analysis (Shortcut, Format Approach):

**Fact 1:** Mesh Analysis will have 2 polarities when shared by 2 loops. Useful for networks with voltage sources.

**Fact 2:** For polarity, the current's "upstream" end of a resistor is always +, and the "downstream" end is -

**Fact 3:** If there is a current source, remove the current source. The resulting path becomes a **supermesh** current. Setup 1 KVL equation for the supermesh and another equation relating the relevant currents equal to the current source.

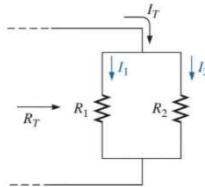
1. Assign CW current loop ( $I_1, I_2$ , etc.) at each "window," or closed loop of a network.
2. Indicate polarities determined by the direction of the loop current.
3. Setup blank  $n \times n$  matrix where  $n = \#$  of nodes for current.
4. Find resistance matrix elements, polarity is not considered.
  - Diagonal elements:  $R_{ii} = \Sigma R_{\text{mesh}}$
  - Other elements:  $R_{ij} = -R$  shared by loop i and j (make sure it's negative)
  - Note:  $R_{ij} = R_{ji}$
5. [Matrix][currents] = [loop total voltage], then rref

**Voltage Divider** (To find V through a resistor)

$$V_{\text{out}} = V_{\text{in}} \left[ \frac{R_2}{R_1 + R_2} \right] \quad V_1 = \frac{R_1 E}{R_T} = \frac{R_1 E}{R_1 + R_2 + R_3}$$

**Current Divider** (To find I through a resistor)

$$I_x = \frac{R_T}{R_x} I_T \quad I_x = \frac{R_1}{R_1 + R_2} I_T$$



**To Find Voltage, Current too ig**

Nodal Analysis:

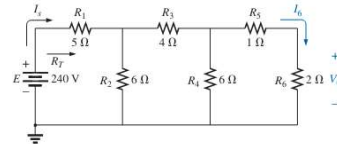
**Fact 1:** Useful for networks with current sources

**Fact 2:** If there is a voltage source, replace the voltage source and any component connected to it with short circuit equivalents and apply KCL to the defined nodes.

Nodal Analysis (Shortcut, Format Approach):

1. Place reference node at known V (usually at ground where  $V=0V$ )
2. Assign voltage to essential nodes ( $V_1, V_2$ , etc.)

3. Setup blank  $n \times n$  matrix where  $n = \#$  of nodes for voltage.
4. Apply KCL with only the current sources directly connected to the node for the RHS of matrix (aka 6A in, 4A out,  $V_1 = 2A$ )
5. Find conductance matrix elements using resistors
  - Diagonal Elements:  $G_{ii} = \text{sum of } G \text{ connected to each node}$
  - Other elements:  $G_{ij} = -G$  of resistor between 2 node i and node j (make sure it's negative!)
  - Note:  $G_{ij} = G_{ji}$
6. [Matrix][voltages] = [node current], then rref



**Ladder Networks I & V**

1. Reduce network to find  $R_T$  and  $I_T$  (4 circo)
2. Find  $I_3$  using  $I_3 = \frac{R_4}{R_5 + R_6} I_T$  (refer circ 3)
3. Find  $I_6$  using  $I_6 = \frac{R_5 + R_6}{R_4 + R_5 + R_6} I_3$  (refer circ 2)
4.  $V_6 = I_6 R_6$  (refer circ 1)

**Series Circuits**

Current in Series:  $I_S = I_1 = I_2 = I_3$

Total Current in Series:  $I_S = \frac{V}{R_1 + R_2 + R_3} = \frac{V}{R_T}$

Total Resistance in Series:  $R_T = R_1 + R_2 + R_3$

Total Resistance in Series:  $R_T = \frac{E}{I}$

Voltage Sources in Series:  $E_T = E_1 + E_2 + E_3$

**Parallel Circuits**

Voltage in Parallel:  $V_S = V_1 = V_2 = V_3$

Total Resistance in Parallel:  $R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}}$

Conductance:  $G = G_1 + G_2 + \dots + G_N$ , where  $G = 1/R$

Total Current in Parallel:  $I_S = \frac{V}{\frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}}}$

**Max Power of Resistor**

$$P_{\text{max}} = \frac{E_{Th}^2}{4R_{Th}}$$

Remove V

Find  $R_{Th}$

- 1.)  $E_1$  no I
- 2.)  $E_2$  no V

$E_{Th} = E_1 + E_2$



## 11-1 – Electric Field

**TABLE 2.1**  
Relative conductivity of various materials.

| Metal      | Relative Conductivity (%) |
|------------|---------------------------|
| Silver     | 105                       |
| Copper     | <b>100</b>                |
| Gold       | 70.5                      |
| Aluminum   | 61                        |
| Tungsten   | 31.2                      |
| Nickel     | 22.1                      |
| Iron       | 14                        |
| Constantan | 3.52                      |
| Nichrome   | 1.73                      |
| Calorite   | 1.44                      |

$$E = \frac{kQ}{r^2} = \frac{V}{d}$$

$$\text{Capacitance: } C = \frac{Q}{V}$$

$$C = 8.85 \times 10^{-12} \epsilon_r \frac{A}{d}$$

**TABLE 10.1**  
Relative permittivity (dielectric constant)  $\epsilon_r$  of various dielectrics.

| Dielectric                          | $\epsilon_r$ (Average Values) |
|-------------------------------------|-------------------------------|
| Vacuum                              | 1.0                           |
| Air                                 | 1.0006                        |
| Teflon®                             | 2.0                           |
| Paper, paraffined                   | 2.5                           |
| Rubber                              | 3.0                           |
| Polystyrene                         | 3.0                           |
| Oil                                 | 4.0                           |
| Mica                                | 5.0                           |
| Porcelain                           | 6.0                           |
| Bakelite®                           | 7.0                           |
| Aluminum oxide                      | 7                             |
| Glass                               | 7.5                           |
| Tantalum oxide                      | 30                            |
| Ceramics                            | 20–7500                       |
| Barium-strontium titanite (ceramic) | 7500.0                        |

## 11-1 – Transient Response

$$\tau = RC$$

$$V_c = E(1 - e^{t/\tau})$$

$$i_c = \frac{E}{R}(e^{t/\tau})$$

## 11-2 – Discharge

$$V_c(t) = Ee^{-t/\tau}$$

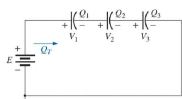
$$\tau = RC$$

$$V_R = Ee^{-t/\tau}$$

$$i_c = \frac{E}{R}e^{t/\tau}$$

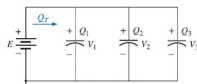
$$i_c = C \frac{dV_c}{dt}$$

## 11-2 – Series/Parallel Capacitor



$$Q_T = Q_1 = Q_2 = Q_3$$

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$



$$Q_T = Q_1 + Q_2 + Q_3$$

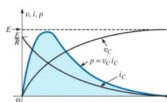
$$C_T = C_1 + C_2 + C_3$$

$$Q_T = C_T E$$

$$V_C = Q_T / C_T$$

## 11-2 – Energy Stored by Capacitor

An ideal capacitor does not dissipate any of the energy supplied to it. It stores the energy in the form of an electric field between the conducting surfaces. The energy stored is represented by the shaded area under the power curve.



$$W_C = \frac{1}{2} CV^2$$

$$W_C = \frac{Q^2}{2C}$$

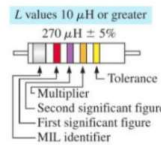
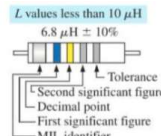
**Fact 1:** W units is Joules (J)

**Fact 2:** Short capacitors for V

## 13-1 – Inductor

| Color Code Table   |                    |                         |                          |
|--------------------|--------------------|-------------------------|--------------------------|
| Color <sup>1</sup> | Significant Figure | Multiplier <sup>2</sup> | Inductance Tolerance (%) |
| Black              | 0                  | 1                       |                          |
| Brown              | 1                  | 10                      |                          |
| Red                | 2                  | 100                     |                          |
| Orange             | 3                  | 1000                    |                          |
| Yellow             | 4                  |                         |                          |
| Green              | 5                  |                         |                          |
| Blue               | 6                  |                         |                          |
| Violet             | 7                  |                         |                          |
| Gray               | 8                  |                         |                          |
| White              | 9                  |                         |                          |
| None <sup>2</sup>  |                    |                         | ±20                      |
| Silver             |                    |                         | ±10                      |
| Gold               |                    |                         | ±5                       |
|                    | Decimal point      |                         |                          |

<sup>1</sup> Indicates body color.  
<sup>2</sup> The multiplier is the factor by which the two significant figures are multiplied to yield the nominal inductance value.



$$L = \mu \frac{N^2 A}{l}$$

$$\text{Permeability: } \mu = \mu_r \mu_0 \left[ \frac{Wb}{A} \cdot m \right]$$

$$\mu_0 = 4\pi \times 10^{-7}$$

N: Number of turns, A: m<sup>2</sup>, l: m, L: Henry

## 13-1 – Inductor Characteristics

### Equation

$$V_L = Ee^{-\frac{t}{\tau}}$$

$$V_L = Ee^{-\frac{t}{\tau}}$$

$$i_L = \frac{E}{R}(1 - e^{-t/\tau})$$

$$V_R = E(1 - e^{-t/\tau})$$

$$\tau = \frac{L}{R}$$

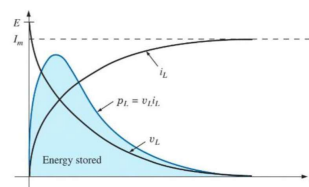
## 13-1 – Initial Condition

$$i_L = I_f + (I_f - I_i)e^{t/\tau}$$

## 13-2 – R-L Transients

$$V_L = L \frac{di_L}{dt}$$

## 13-2 – Inductors Series and Parallel

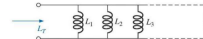
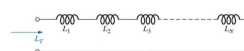


$$W_{\text{stored}} = \frac{1}{2} LI^2$$

$$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_N}$$

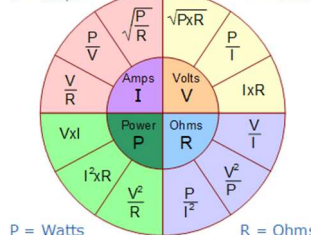
For two inductors in parallel,

$$L_T = \frac{L_1 L_2}{L_1 + L_2}$$



I = Amps

V = Volts



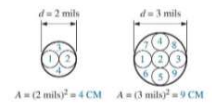
P = Watts

R = Ohms

**TABLE 3.6**  
Temperature coefficient of resistance for various conductors at 20°C.

| Material   | Temperature Coefficient ( $\alpha_{20}$ ) |
|------------|-------------------------------------------|
| Silver     | 0.0038                                    |
| Copper     | <b>0.00393</b>                            |
| Gold       | 0.0034                                    |
| Aluminum   | 0.00391                                   |
| Tungsten   | 0.005                                     |
| Nickel     | 0.006                                     |
| Iron       | 0.0055                                    |
| Constantan | 0.00008                                   |
| Nichrome   | 0.00044                                   |

- One mil is a thousandth of an inch.  
1000 mil = 1 in
- A circle with a diameter of one mil has an area of one Circular Mil (CM)  
 $A_{CM} = (d_{mil})^2$



$$A = \frac{\pi d^2}{4} = \frac{\pi}{4} (1 \text{ mil})^2 = \frac{\pi}{4} \text{ sq mils} = 1 \text{ CM}$$

**TABLE 3.2**  
American Wire Gage (AWG) sizes.

| AWG #             | Area (CM) | $\Omega/1000 \text{ ft}$ at 20°C | Maximum Allowable Current for RHW Insulation (A) <sup>a</sup> |
|-------------------|-----------|----------------------------------|---------------------------------------------------------------|
| (4/0) <b>0000</b> | 211,600   | 0.0490                           | <b>230</b>                                                    |
| (3/0) <b>000</b>  | 167,810   | 0.0618                           | <b>200</b>                                                    |
| (2/0) <b>00</b>   | 133,080   | 0.0780                           | <b>175</b>                                                    |
| (1/0) <b>0</b>    | 105,530   | 0.0983                           | <b>150</b>                                                    |
| <b>1</b>          | 83,694    | 0.1240                           | <b>130</b>                                                    |
| <b>2</b>          | 66,373    | 0.1563                           | <b>115</b>                                                    |
| <b>3</b>          | 52,634    | 0.1970                           | <b>100</b>                                                    |
| <b>4</b>          | 41,742    | 0.2485                           | <b>85</b>                                                     |
| <b>5</b>          | 33,102    | 0.3133                           | —                                                             |
| <b>6</b>          | 26,250    | 0.3951                           | <b>65</b>                                                     |
| <b>7</b>          | 20,816    | 0.4982                           | —                                                             |
| <b>8</b>          | 16,509    | 0.6282                           | <b>50</b>                                                     |
| <b>9</b>          | 13,094    | 0.7921                           | —                                                             |
| <b>10</b>         | 10,381    | 0.9989                           | <b>30</b>                                                     |
| <b>11</b>         | 8,234.0   | 1.260                            | —                                                             |
| <b>12</b>         | 6,529.9   | 1.588                            | <b>20</b>                                                     |
| <b>13</b>         | 5,178.4   | 2.003                            | —                                                             |
| <b>14</b>         | 4,106.8   | 2.525                            | <b>15</b>                                                     |
| <b>15</b>         | 3,256.7   | 3.184                            | —                                                             |
| <b>16</b>         | 2,582.9   | 4.016                            | —                                                             |
| <b>17</b>         | 2,048.2   | 5.064                            | —                                                             |
| <b>18</b>         | 1,624.3   | 6.385                            | —                                                             |
| <b>19</b>         | 1,288.1   | 8.051                            | —                                                             |
| <b>20</b>         | 1,021.5   | 10.15                            | —                                                             |
| <b>21</b>         | 810.10    | 12.80                            | —                                                             |
| <b>22</b>         | 642.40    | 16.14                            | —                                                             |
| <b>23</b>         | 509.45    | 20.36                            | —                                                             |
| <b>24</b>         | 404.01    | 25.67                            | —                                                             |
| <b>25</b>         | 320.40    | 32.37                            | —                                                             |
| <b>26</b>         | 254.10    | 40.81                            | —                                                             |
| <b>27</b>         | 201.50    | 51.47                            | —                                                             |
| <b>28</b>         | 159.79    | 64.90                            | —                                                             |
| <b>29</b>         | 126.72    | 81.83                            | —                                                             |
| <b>30</b>         | 100.50    | 103.2                            | —                                                             |
| <b>31</b>         | 79.70     | 130.1                            | —                                                             |
| <b>32</b>         | 63.21     | 164.1                            | —                                                             |
| <b>33</b>         | 50.13     | 206.9                            | —                                                             |
| <b>34</b>         | 39.75     | 260.9                            | —                                                             |
| <b>35</b>         | 31.52     | 329.0                            | —                                                             |
| <b>36</b>         | 25.00     | 414.8                            | —                                                             |
| <b>37</b>         | 19.83     | 523.1                            | —                                                             |
| <b>38</b>         | 15.72     | 659.6                            | —                                                             |
| <b>39</b>         | 12.47     | 831.8                            | —                                                             |
| <b>40</b>         | 9.89      | 1049.0                           | —                                                             |

**TABLE 3.3**  
Resistivity ( $\rho$ ) of various materials.

| Material  | $\Omega\text{-cm}$                       |
|-----------|------------------------------------------|
| Silver    | $1.645 \times 10^{-6}$                   |
| Copper    | <b><math>1.723 \times 10^{-6}</math></b> |
| Gold      | $2.443 \times 10^{-6}$                   |
| Aluminum  | $2.825 \times 10^{-6}$                   |
| Tungsten  | $5.485 \times 10^{-6}$                   |
| Nickel    | $7.811 \times 10^{-6}$                   |
| Iron      | $12.299 \times 10^{-6}$                  |
| Tantalum  | $15.54 \times 10^{-6}$                   |
| Nichrome  | $99.72 \times 10^{-6}$                   |
| Tin oxide | $250 \times 10^{-6}$                     |
| Carbon    | $3500 \times 10^{-6}$                    |

**TABLE 3.5**  
Inferred absolute temperatures ( $T_i$ ).

| Material   | °C            |
|------------|---------------|
| Silver     | −243          |
| Copper     | <b>−234.5</b> |
| Gold       | −274          |
| Aluminum   | −236          |
| Tungsten   | −204          |
| Nickel     | −147          |
| Iron       | −162          |
| Nichrome   | −2,250        |
| Constantan | −125,000      |